

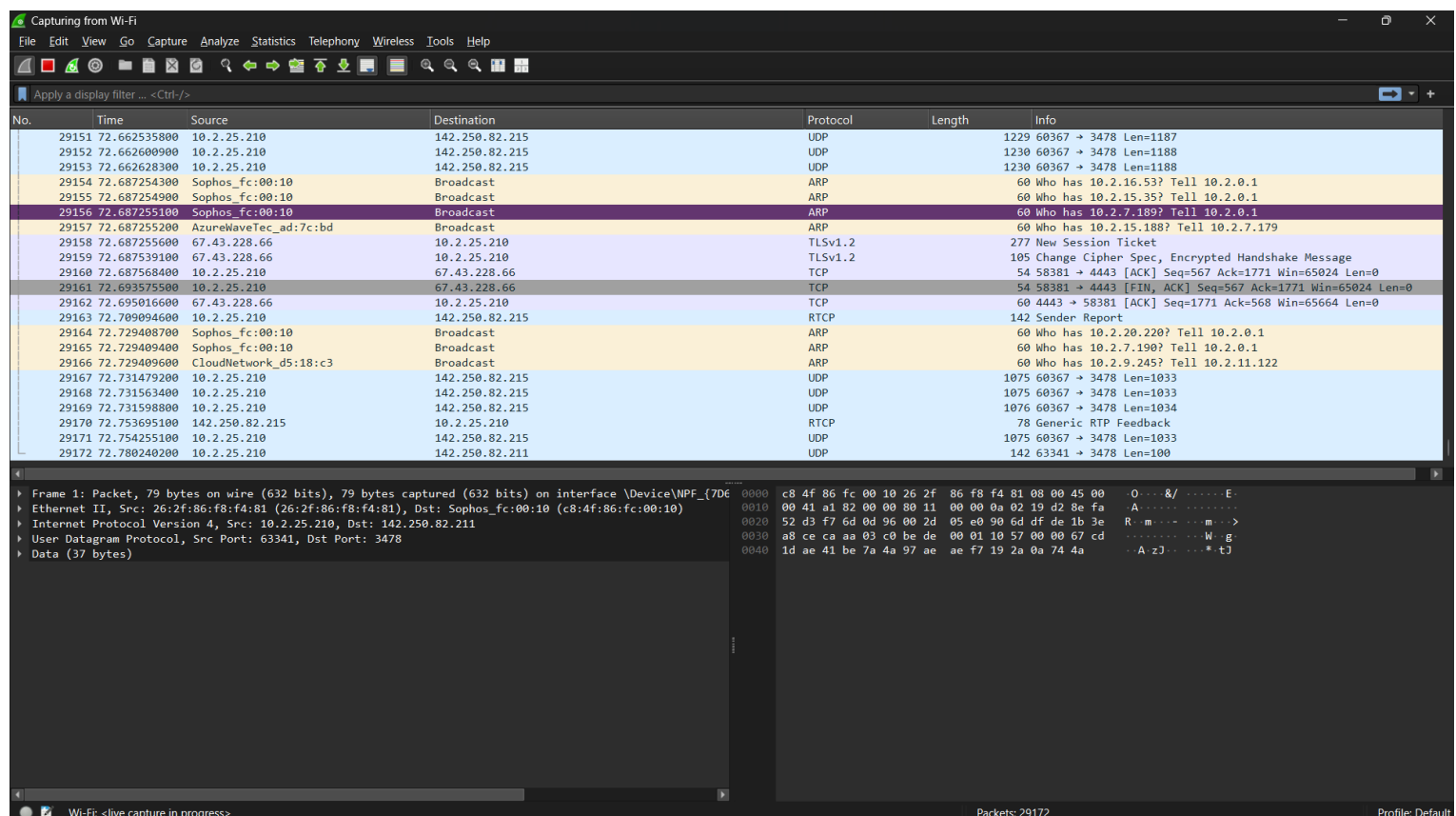
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.

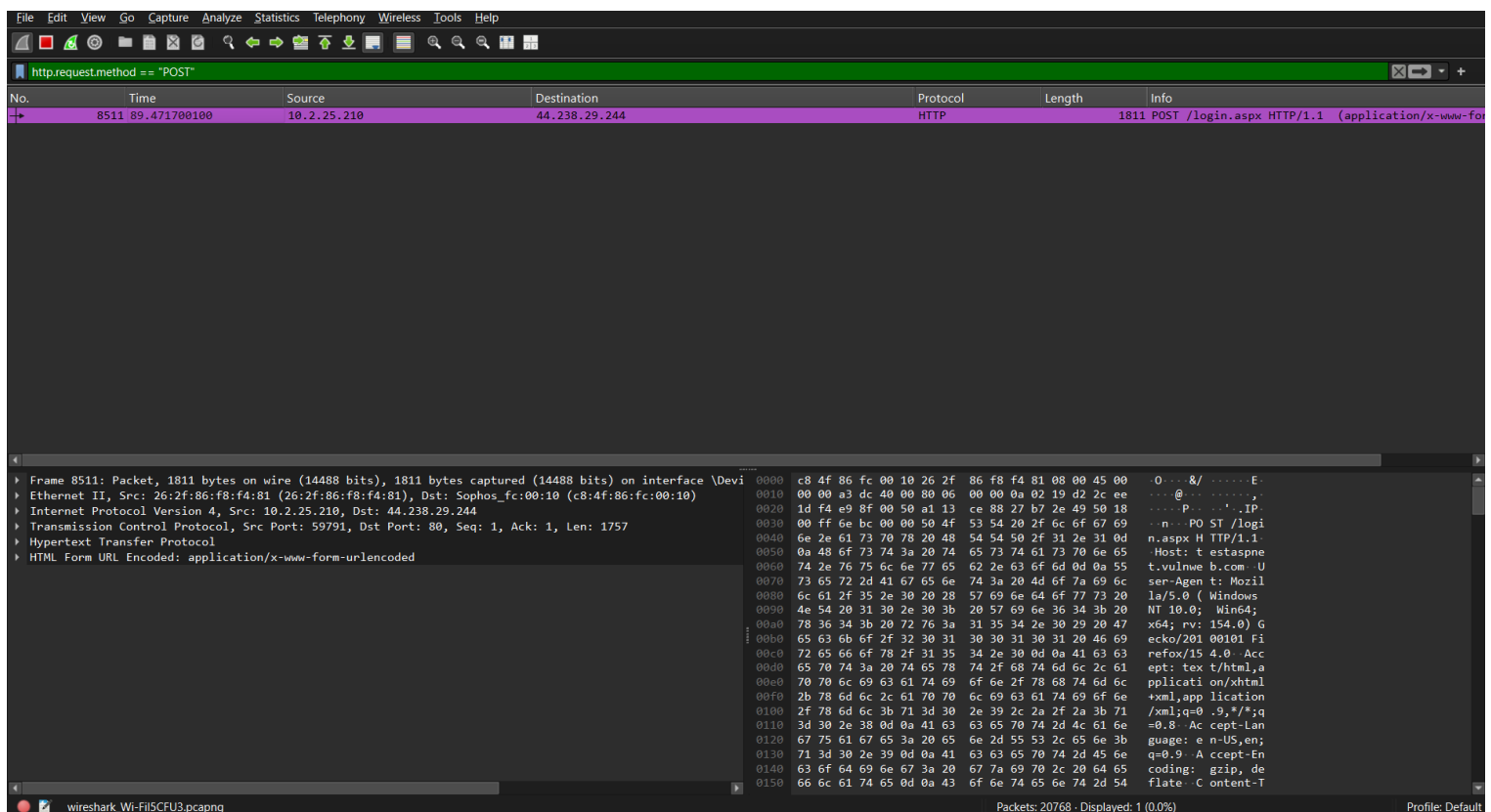


Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image.



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can see in the below image the green bar where we apply the filter.
http



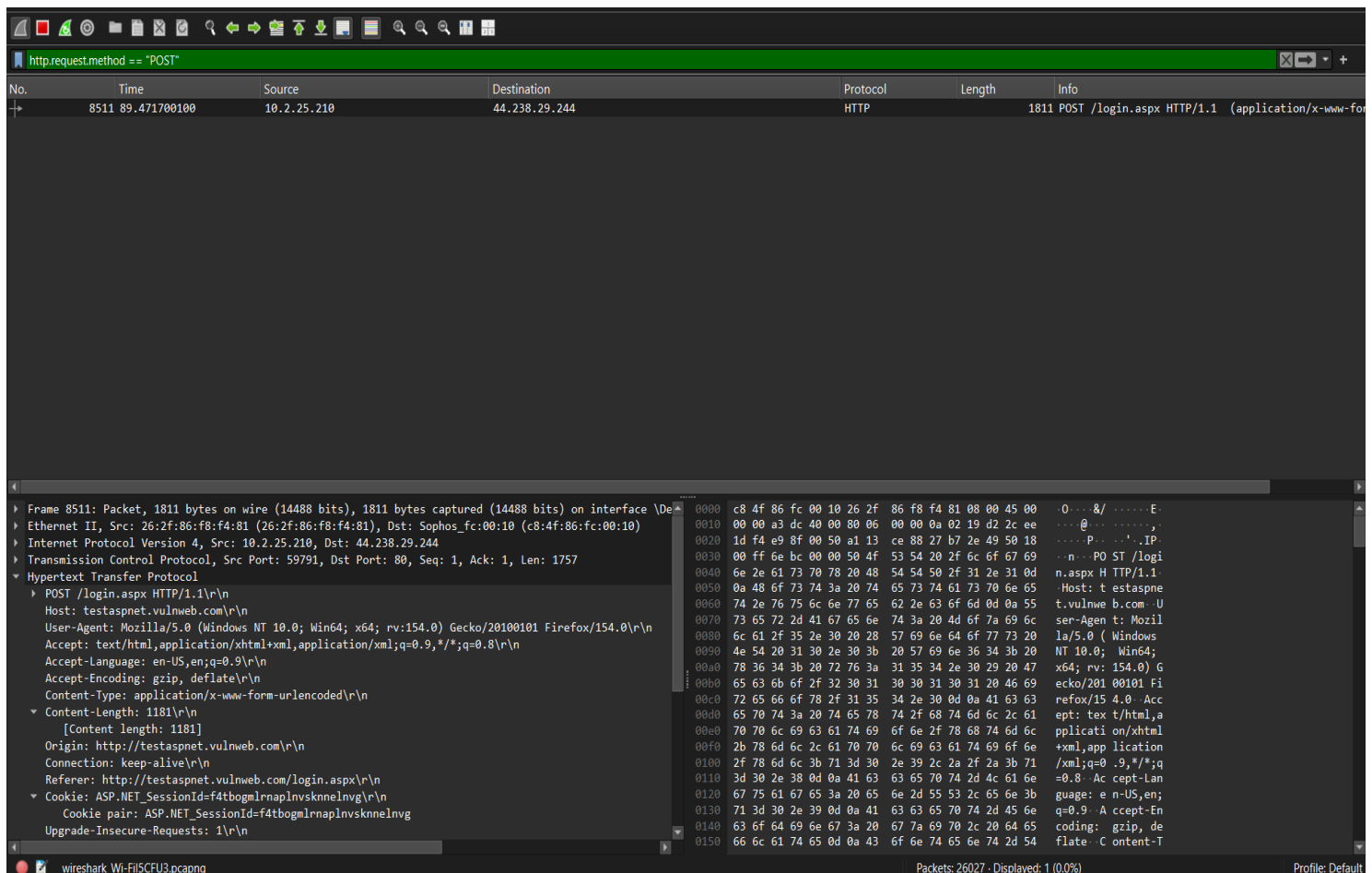
Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-

- GET
- POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

`http.request.method == "GET"`



GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

`http.request.method == "POST"`

http.request.method == "POST"						
No.	Time	Source	Destination	Protocol	Length	Info
8511	89.471700100	10.2.25.210	44.238.29.244	HTTP	1811	POST /login.aspx HTTP/1.1 (application/x-www-form-urlencoded)

Frame 8511: Packet, 1811 bytes on wire (14488 bits), 1811 bytes captured (14488 bits) on interface \Device\NPF{...}

Ethernet II, Src: 26:2f:86:f8:f4:81 (26:2f:86:f8:f4:81), Dst: Sophos_fc:00:10 (c8:4f:86:fc:00:10)

Internet Protocol Version 4, Src: 10.2.25.210, Dst: 44.238.29.244

Transmission Control Protocol, Src Port: 59791, Dst Port: 80, Seq: 1, Ack: 1, Len: 1757

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

Form item: "__EVENTTARGET" = ""

Form item: "__EVENTARGUMENT" = ""

[...] Form item: "__VIEWSTATE" = "/wEPDwUKLTiYmzk2OTgxMQ9kFgICAQ9kFgICAQ9kFgQCAQ8wBB4EaH1ZgUKbG9naW4uYw=="

Form item: "__VIEWSTATEGENERATOR" = "C2EE9AB8"

[...] Form item: "__EVENTVALIDATION" = "/wEWwWLoz/fGCGlStq24BwK3jsrkBALtufvLDQKC3IeGDAKR1ca+DALi++jXCAL"

Form item: "tbUsername" = "exp3"

Form item: "tbPassword" = "123456"

Form item: "cbPersistCookie" = "on"

Form item: "btnLogin" = "Login"

0000 c8 4f 86 fc 00 10 26 2f 86 f8 f4 81 08 00 45 00 .O....&/.....E.

0010 00 00 a3 dc 40 00 80 06 00 00 0a 02 19 d2 2c ee@.....,

0020 1d f4 e9 8f 00 50 a1 13 ce 88 27 b7 2e 49 50 18P.....IP.

0030 00 ff 6e bc 00 00 50 4f 53 54 20 2f 6c 6f 67 69n...PO ST /logi

0040 6e 2e 61 73 70 78 20 48 54 54 50 2f 31 2e 31 0d n.aspx H TTP/1.1

0050 0a 48 6f 73 74 3a 20 74 65 73 74 61 73 70 6e 65 .Host: t estaspne

0060 74 2e 76 75 6c 6e 77 65 62 2e 63 6f 6d 0d 0a 55 t.vulnwe b.com .U

0070 73 65 72 2d 41 67 65 6a 74 3a 20 4d 6f 7a 69 6c ser-Agent t: Mozil

0080 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 la/5.0 (Windows

0090 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 NT 10.0; Win64;

00a0 78 36 34 3b 20 72 76 3a 31 35 34 2e 30 29 20 47 x64; rv: 154.0) G

00b0 65 63 6b 6f 2f 32 30 31 30 30 31 30 31 20 46 69 ecko/201 00101 Fi

00c0 72 65 66 6f 78 2f 31 35 34 2e 30 0d 0a 41 63 63 refox/15 4.0 . Acc

00d0 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 ept: tex t/html,a

00e0 70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c pplicati on/xhtml

00f0 2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e +xml,app lication

0100 2f 78 6d 6c 3b 71 3d 30 2e 39 2c 2a 2f 2a 3b 71 /xml;q=0 .9,*/*;q

0110 3d 30 2e 38 0d 0a 41 63 65 70 74 2d 4c 61 6e =0.8 .Ac cept-Lan

0120 67 75 61 67 65 3a 20 65 6e 2d 55 53 2c 65 6e 3b guage: e n-US,en;

0130 71 3d 30 2e 39 0d 0a 41 63 65 70 74 2d 45 6e q=0.9 .A ccept-En

0140 63 6f 64 69 6e 67 3a 20 67 7a 69 70 2c 20 64 65 coding: gzip, de

0150 66 6c 61 74 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 flate C ontent-T

wireshark Wi-FiScanner.pcapng

Packets: 27113 · Displayed: 1 (0.0%)

Profile: Default

As you can see we have a packet with form data click on the packet with user info and the application URL encoded. and click on the down-HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "uname" = "exp3" Form item: "pass" = "123456"